

Lecture Notes on

Condensed Matter

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cond-mat

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1 Lattices

1.1 Lattice structures

The atoms in most solids arrange themselves in periodically repeating structures called *lattices*. Formally we'll say that an m -dimensional lattice Λ is a $\mathbb{Z}$ -module generated by some linearly independent subset of vectors $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^n$. We call these vectors *primitive lattice vectors*. The primary example to keep in mind is the *square lattice*, which is the lattice generated by the standard basis vectors, $\mathbf{a}_i = \mathbf{e}_i$.

Given a lattice $\Lambda \subset \mathbb{R}^n$, we can mod out by the lattice structure to implement the physical equivalence

$$\mathbf{x} \sim \mathbf{x} + \sum_i n_i \mathbf{a}_i.$$

This gives us the “fundamental domain” of our space, i.e. the space of actually novel motions in it. We'll denote this as $X = \mathbb{R}^n / \Lambda$. This will be useful later when we talk about functions on the lattice.

We can consider portions of our space at a time—we call these subsets of $\mathbb{R}^n$ *cells*. There are three kinds of cells that are disproportionately useful to consider. First, notice that if we consider X as a subset of $\mathbb{R}^n$, by translating it around with Λ , we fill the entire space $\mathbb{R}^n$. Said differently, the orbit of X , $\mathcal{O}_X = \mathbb{R}^n$. We will call cells of this type *unit* cells. We will call the unit cell with the fewest generators a *primitive* cell. Lastly, there is a special kind of unit cell that naturally manifests the symmetries of the lattice Λ called a *Wigner-Seitz* cell. It's defined as

$$\Gamma = \{\mathbf{x} \in \mathbb{R}^n \mid |\mathbf{x}| < |\mathbf{x} - \mathbf{A}| \text{ for all } \mathbf{A} \in \Lambda\}.$$

That is: define some arbitrary point in your lattice as the origin and consider the closest elements to it such that translating the cell around fills the entire space—this is the Wigner-Seitz cell. We can see that this cell naturally manifests the symmetries of the lattice by considering the group action of $\Lambda \curvearrowright \Gamma$. Letting $g \in \Lambda$ act on the inequality above, we have

$$|g\mathbf{x}| < |g\mathbf{x} - g\mathbf{A}| \text{ for all } \mathbf{A} \in \Lambda.$$

But, $g\mathbf{A}$ runs over the elements of Λ when $\mathbf{A}$ does, and thus if $\mathbf{x} \in \Gamma$, then

$$|g\mathbf{x}| < |g\mathbf{x} - \mathbf{A}| \text{ for all } \mathbf{A} \in \Lambda.$$

That is: $g\mathbf{x} \in \Gamma$ as well, so $g\Gamma = \Gamma$, i.e. the Wigner-Seitz cell only rearranges itself under the lattice symmetries.

There are a few natural constructions we can now take stock of. The first is to note that we can easily define the lattice dual in the standard way,

$$\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{R}).$$

We can define a basis for $\Lambda^\vee$ in natural way, i.e. by considering the set of functionals $\varphi \in \Lambda^\vee$ dual to the primitive lattice vectors,

$$\varphi^i(\mathbf{a}_j) = \delta_j^i.$$

Furthermore, since $\mathbb{R}^n$ has an inner product on it, we can associate to each functional φ^i a unique vector $\mathbf{g}_i \in \Lambda$ such that for any $\mathbf{A} \in \Lambda$,

$$\varphi^i(\mathbf{A}) = \mathbf{g}_i \cdot \mathbf{A}.$$

This is a natural consequence of the isomorphism $\Lambda \cong \Lambda^\vee$. For our purposes, it will be more useful to rescale the reciprocal basis vectors by 2π ; that is, instead choose a basis of $\Lambda^\vee$ such that

$$\mathbf{g}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}.$$

$\Lambda^\vee$ itself can be thought of as a lattice due to the isomorphism $\Lambda^\vee \cong \Lambda$, and so all of the previous discussion about cells goes through the same. We call the Wigner-Seitz cell for $\Lambda^\vee$ the *Brillouin zone*, $\mathcal{B}$, and it's one of the most important constructions in condensed matter.

For example, if $\Lambda = \langle a \rangle$, then the Wigner-Seitz cell for the lattice is $\Gamma = [-a/2, a/2]$, and the Brillouin zone is $\mathcal{B} = [-\pi/a, \pi/a]$. This is easily seen by noting that $\Lambda^\vee = \langle g = 2\pi/a \rangle$, as $ga = 2\pi$, and so $\mathcal{B} = [-g/2, g/2]$ —an line of diameter g centered at the origin.

1.2 Constructions on the lattice

We now define functions $\psi(\mathbf{x})$ on $\mathbb{R}^n$ with an embedded lattice $\Lambda \hookrightarrow \mathbb{R}^n$. It's intuitively clear that if we describe $\psi(\mathbf{x})$ on $X = \mathbb{R}^n/\Lambda$, then by translational symmetry we've described it on all of $\mathbb{R}^n$.

However, all of the above discussion has implicitly been assuming a free particle $\psi(\mathbf{x})$ —*Bloch's theorem* gives us insight into what changes when we add a potential $V(\mathbf{x})$ that respects the lattice structure. Let $V(\mathbf{x}) = V(\mathbf{x} + \mathbf{A})$, where $\mathbf{A} \in \Lambda$; that is, V is a periodic potential on the lattice. Then Bloch's theorem states that we can label energy eigenstates purely by their crystal momentum $\mathbf{k}$:

$$\psi_{\mathbf{k}}(\mathbf{x}) = e^{i\mathbf{k} \cdot \mathbf{x}} u_{\mathbf{k}}(\mathbf{x}),$$

where $u_{\mathbf{k}}(\mathbf{x})$ is a function periodic in Λ . We have been implicitly working in the *extended zone scheme* for $\mathbf{k}$. This is where we do not implement the equivalence $\mathbf{k} \sim \mathbf{k} + \mathbf{G}$, with $\mathbf{G} \in \Lambda^\vee$, and so $\mathbf{k}$ ranges through all of $\mathbb{R}^n$. We may also explicitly implement this equivalence and work in the *reduced zone scheme*, where we let $\mathbf{k} \in \mathcal{B}$ instead, and then add an extra index to ψ to resolve degeneracies:

$$\psi_{\mathbf{k},n}(\mathbf{x}) = e^{i\mathbf{k} \cdot \mathbf{x}} u_{\mathbf{k},n}(\mathbf{x}).$$

The way to prove this is as follows: consider a representation of Λ , $\rho_\Lambda: \Lambda \rightarrow GL(\mathcal{H})$ by $\mathbf{A} \mapsto T_{\mathbf{A}}$, where $\mathcal{H}$ is our Hilbert space and $\mathbf{A} \in \Lambda$. These are *translation operators* $T_{\mathbf{A}}$ acting on ψ as

$$T_{\mathbf{A}}\psi(\mathbf{x}) = \psi(\mathbf{x} + \mathbf{A}).$$

It is trivial to verify that the T 's form a representation of Λ . Note that since Λ is abelian, $T_{\mathbf{A}} = e^{i\theta_{\mathbf{A}}}$ for some $\theta_{\mathbf{A}} \in [0, 2\pi)$.

Since the wave functions are symmetric under translations by Λ , the translation operators commute with the Hamiltonian, $[T_{\mathbf{A}}, H] = 0$. Thus, H and $T_{\mathbf{A}}$ are simultaneously diagonalizable. We now have to consider what the natural quantum number is to label our wave-functions with. Recall that, since translation is a commutative operation, the translation operators must act like a phase. Furthermore, since $\psi(\mathbf{x}) = \psi(\mathbf{x} + \mathbf{A})$ for any $\mathbf{A} \in \Lambda$, we must have

$$\psi(\mathbf{x}) = \psi(\mathbf{x} + \mathbf{A}) = T_{\mathbf{A}}\psi(\mathbf{x}) = e^{i\theta_{\mathbf{A}}}\psi(\mathbf{x}) \implies \theta_{\mathbf{A}} = 2\pi n.$$

Thus, we can write $\theta_{\mathbf{A}} = \mathbf{k} \cdot \mathbf{A}$, where $\mathbf{k} \in \Lambda^\vee$, as $\mathbf{k} = \sum_i a_i \mathbf{g}_i$, and $\mathbf{g}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$. So, we can label our states by the quantum number $\mathbf{k} \in \mathcal{B}$. Bloch's theorem follows when one considers the definition of $u_{\mathbf{k}}(\mathbf{x}) = e^{-i\mathbf{k} \cdot \mathbf{x}} \psi_{\mathbf{k}}(\mathbf{x})$ and implements the periodicity of the lattice structure.